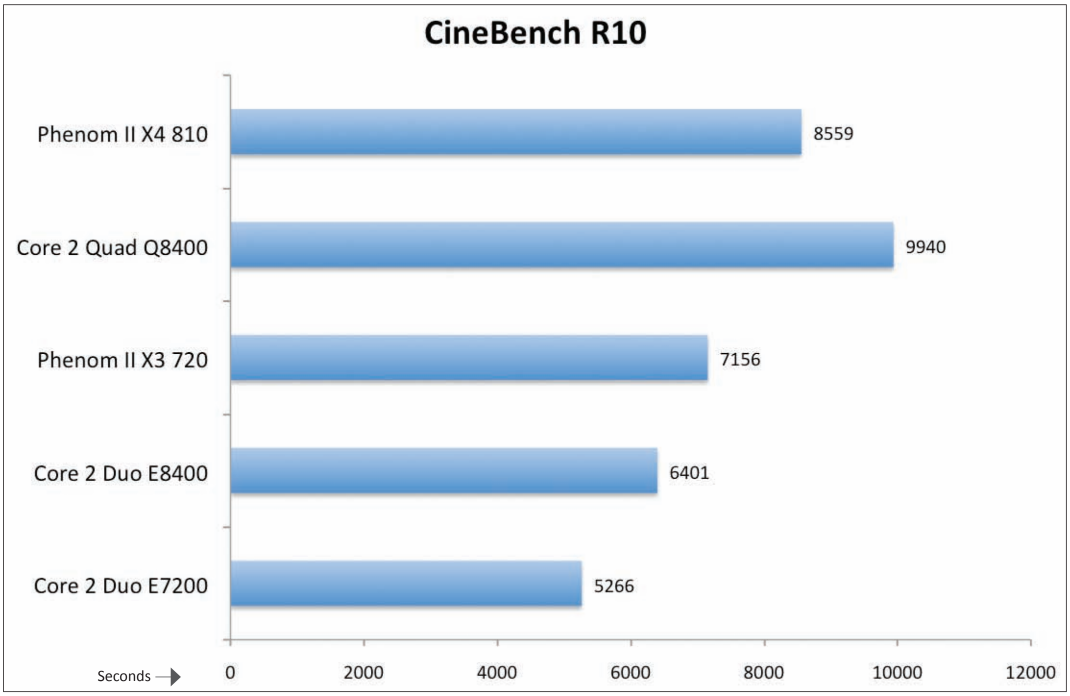
Cheaper Graphene?

Three professors from Northwestern's McCormick School of Engineering and Applied Science have created a cheaper technique of manufacturing graphene, which is touted as a possible silicon replacement

IE8 a tortoise?

FutureMark Corp. said IE8 was the slowest browser around and even Safari from Apple was much faster

Phenom II X2 550BE and the Intel E8400 best this test. If anything, the E8400 is the faster of the two despite having a 100 MHz clock disadvantage. *Tales of Valor* also favours the fast E8400. However, the video encoding benchmarks, WinRAR 3.9 and Maya 2008, all of which are heavily multithreaded, favour the quad-core processors. In fact, in the HandBrake x264 encoding test the Q8400 was nearly twice as fast as the E8400. Encoding applications are SSE-based and generally make better use of multiple cores. Intel's quad cores are cheaper than AMD's, a fact that we immediately pounced on. The Q8400 was our Best Buy winner as a result of this and it's superb all round performance. If you



EDITOR'S PICK digit SEPTEMBER 2009

AMD Athlon X2 550

are interested in building a powerful PC, this processor is very suitable. If all you are interested in doing is gaming then we recommend the E8400. However, at just a couple of hundred rupees less, it doesn't offer much more. We think all the CPUs in this comparison are more than suitable for reasonably powerful all-purpose computers. For a rendering workstation, give preference to quad core processors. If you have a lower budget in the range of Rs. 6,500, you can't go wrong with the Phenom II X2 550BE – at Rs. 6,150, it's great value. As excellent as the Phenom II X4 810 and X4 905E CPUs are, we cannot recommend them on account of the Core 2 Quad Q8400 – if you are considering spending anything like Rs. 9,000 for a CPU, you'd have to be daft to look elsewhere.

Processors above Rs. 12,500 – Taking no prisoners

Shopping for a processor in excess of Rs. 12,500 is a fun experience. Well, shopping with deep pockets anywhere is fun isn't it? When spending this much, be prepared to pair your CPU with a feature rich, high-end chipset-based motherboard. Also be prepared to invest in at least 4 GB of RAM, otherwise you'll end up with a hungry, sporadically fed beast. Some of the processors in this category are extremely expensive, and to be fair the reason warrants mention. We're talking about Intel's Extreme series of processors. Traditionally, Intel has always charged a pretty premium for anything it deems as *Extreme Edition*. These processors are little more than the highest clocked parts with upward unlocked multipliers. Intel also claims that these CPUs are made of the best binned components and the highest yield silicon – which is why they are better suited to stand the rigours of over-clocking. To some benchmark gurus, these CPUs are a godsend allowing them to break all sorts of world records. To the more mundane person spending upward of Rs. 50,000 on a CPU, then over-clocking it, and thereby risking its functionality, may seem plumb crazy – however, that is the way it stands.

It's interesting to note that AMD has been hard at work ramping up clock speeds for its CPUs. The AMD Phenom II 965BE clocks at a mighty 3.4 GHz, and AMD has already announced the Phenom II 975BE rated at 3.6 GHz – whoever said the gigahertz wars have ended must have been on weed – the fact is that the battleground just changed. Another worthwhile point is the temperature of these processors. Throughout this test we monitored CPU temperatures but found little concern – all of today's CPUs behave themselves and do not overheat if installed properly. However, AMD's new Phenom II 965BE does get a little

BEST BUY digit SEPTEMBER 2009

Intel Core i7 920